

Customer Shopping Behavior Analysis

End-to-End Data Analytics Project Report

Tools Used: Python • MySQL • Power BI

Dataset: 3,900 customer records • 18 columns

Date: May 2026

1. Project Overview

This project was developed to help a company understand how its customers shop. Using a dataset of 3,900 purchase records, the goal was to uncover patterns in spending behavior, product preferences, customer segments, and subscription trends — all of which directly inform strategic business decisions.

The project followed a three-phase pipeline:

- **Phase 1:** Data cleaning and exploration using Python (Pandas)
- **Phase 2:** Structured SQL analysis using MySQL Workbench
- **Phase 3:** Interactive dashboard development using Power BI

This end-to-end approach demonstrates how raw data can be transformed into actionable insight using industry-standard tools.

2. Dataset Summary

The dataset used in this project was a customer shopping behavior CSV file with the following characteristics:

Attribute	Details
Total Rows	3,900
Total Columns	18
Missing Values	37 values in Review Rating column
Key Demographic Fields	Age, Gender, Location, Subscription Status
Key Purchase Fields	Item Purchased, Category, Purchase Amount, Season, Size, Color
Key Behavioral Fields	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Shipping Type, Payment Method
Engineered Features	age_group, purchase_frequency_days

3. Exploratory Data Analysis (Python)

Python was used as the first step to load, clean, and prepare the dataset before moving to SQL. The libraries used were Pandas and NumPy.

3.1 Data Loading & Initial Inspection

- Imported the dataset using `pandas.read_csv()`

- Used `df.info()` to inspect data types and null counts
- Used `df.describe()` to review statistical summaries (mean, min, max, std)

3.2 Missing Data Handling

- Found 37 missing values in the `review_rating` column
- Imputed missing values using the median rating per product category to preserve data integrity without bias

3.3 Column Standardization

- All column names were converted to `snake_case` for consistency and better readability
- Example: "Purchase Amount (USD)" became `purchase_amount`

3.4 Feature Engineering

- `age_group` column created by binning customer ages into: Young Adult, Adult, Middle-aged, Senior
- `purchase_frequency_days` column created by mapping text values (e.g., Weekly, Monthly) to numeric day values using a dictionary mapping
- Discovered a typo in the mapping key: 'Anually' vs the actual data value 'Annually' — corrected to avoid NaN values

3.5 Data Consistency Check

- Verified that `discount_applied` and `promo_code_used` columns were 100% identical using `(df['discount_applied'] == df['promo_code_used']).all()`
- Result: `np.True_` — confirmed redundancy, so `promo_code_used` was dropped using `df.drop('promo_code_used', axis=1)`

3.6 Database Integration

- Connected Python to MySQL using SQLAlchemy's `create_engine()` function
- Password with special characters (`@`) was handled using `urllib.parse.quote_plus()` to avoid URL parsing errors
- Loaded the cleaned DataFrame into MySQL using `df.to_sql()` with `if_exists='replace'`

4. Data Analysis using MySQL

Once the cleaned data was loaded into MySQL, structured queries were written to answer 10 key business questions. Each query was saved in a separate SQL file tab within MySQL Workbench.

Q#	Business Question	Key Finding
Q1	What is the total revenue by gender?	Male: ₹1,57,890 Female: ₹75,191
Q2	Customers using discounts but spending above average?	839 customers identified
Q3	Top 5 products by average review rating?	Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)
Q4	Average purchase: Express vs Standard shipping?	Express: ₹60.48 Standard: ₹58.46
Q5	Do subscribers spend more?	Subscribers: avg ₹59.49 Non-subscribers: avg ₹59.87 (very similar)
Q6	Top 5 products by discount rate?	Hat (50%), Sneakers (49.7%), Coat (49.1%), Sweater (48.2%), Pants (47.4%)
Q7	Customer segmentation by purchase history?	New / Returning / Loyal segments identified using CASE + CTE
Q8	Top 3 products per category?	Used ROW_NUMBER() window function with PARTITION BY category
Q9	Do repeat buyers subscribe more?	958 repeat buyers subscribed 2,518 did not
Q10	Revenue contribution by age group?	Young Adult: ₹62,143 (highest) Senior: ₹55,763 (lowest)

Key SQL Techniques Used

- Aggregate functions: SUM(), AVG(), COUNT(), ROUND()
- Conditional logic: CASE WHEN ... THEN ... ELSE ... END
- Subqueries: SELECT AVG() used inside WHERE clause
- Common Table Expressions (CTEs): WITH ... AS () for segmentation
- Window functions: ROW_NUMBER() OVER (PARTITION BY ...)
- Filtering & grouping: WHERE, GROUP BY, ORDER BY, LIMIT

5. Power BI Dashboard

The final phase of the project was building an interactive dashboard in Power BI Desktop, connecting directly to the MySQL customer_behavior database.

5.1 Database Connection

- Connected Power BI to MySQL using the MySQL database connector
- Required installation of MySQL Connector/NET before connection was possible
- Entered localhost as server and customer_behavior as the database

- Authenticated using Database credentials (root user)

5.2 Dashboard Visuals

Visual	Type	Insight Shown
No. of Customers	Card	3.9K total customers
Average Review Rating	Card	3.75 out of 5
Average Purchase Amount	Card	\$59.76
% of Customers by Subscription	Donut Chart	73% No, 27% Yes
Revenue by Category	Bar Chart	Clothing leads in revenue
Sales by Category	Bar Chart	Volume comparison across categories
Revenue by Age Group	Bar Chart	Young Adults contribute most
Sales by Age Group	Bar Chart	Consistent across age groups

5.3 Filters & Slicers

- Subscription Status (Yes / No)
- Gender (Female / Male)
- Category (Accessories, Clothing, Footwear, Outerwear)
- Shipping Type (2-Day, Express, Free Shipping, Next Day Air, Standard)

6. Key Findings & Insights

Revenue & Spending

- Male customers generated significantly more revenue than female customers
- Express shipping users spend slightly more on average than Standard shipping users
- Subscribers and non-subscribers spend nearly the same amount on average — subscription status does not predict spending level

Product Performance

- Gloves and Sandals have the highest average review ratings
- Hat, Sneakers, and Coat have the highest discount application rates (around 50%)
- Blouse, Jewelry, and Sandals are the top-selling items within their respective categories

Customer Segments

- Young Adults are the most valuable age group by revenue contribution
- A large proportion of repeat buyers (2,518 out of ~3,476) do not subscribe, indicating a gap between loyalty and subscription conversion
- 839 customers used discounts but still spent above the average — high-value, deal-seeking customers worth targeting

7. Business Recommendations

Recommendation	Rationale
Boost Subscription Campaigns	Only 27% subscribe despite 73% being repeat buyers — targeted campaigns could improve conversion
Customer Loyalty Program	Reward repeat buyers to move them from Returning to Loyal segment and encourage subscription sign-up
Targeted Marketing for Young Adults	This age group generates the most revenue — campaigns should prioritize them
Review Discount Policy	Top-discounted products (Hat, Sneakers) have near 50% discount rate — balance margin with volume
Promote Top-Rated Products	Highlight Gloves, Sandals, and Boots in campaigns as they have the highest satisfaction scores
Express Shipping Upsell	Express users spend more — promote express shipping as a premium, revenue-driving option

8. Tools & Technology Stack

Tool	Purpose	Key Libraries / Features
Python (Jupyter Notebook)	Data cleaning, EDA, feature engineering	pandas, numpy, sqlalchemy, urllib
MySQL Workbench	SQL-based analysis, query execution	CTEs, Window Functions, Aggregations
Power BI Desktop	Interactive dashboard and data visualization	MySQL connector, slicers, card visuals, bar/donut charts
SQLAlchemy	Python-to-MySQL bridge	create_engine(), df.to_sql()

9. Learnings & Takeaways

This project provided hands-on experience across the full data analytics pipeline and reinforced several important concepts:

- Data is rarely clean — small issues like typos in dictionary keys (Anually vs Annually) can silently corrupt entire columns
- SQL is powerful for structured business questions — window functions like `ROW_NUMBER()` enable complex analysis cleanly
- Connecting tools requires understanding of their interfaces — SQLAlchemy URL format, MySQL Connector, Power BI authentication each had unique setup requirements
- Visualization makes findings accessible — the Power BI dashboard turned query results into decisions
- The full pipeline (Python → MySQL → Power BI) mirrors real-world analytics workflows used by data teams in companies

Overall, this project demonstrated that data analytics is not just about writing queries — it is about understanding the business, asking the right questions, and communicating findings clearly.

End of Report